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The Editor-in-Chief
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Dear Editor-in-Chief,

We are pleased to submit our manuscript entitled **"Are Quantum-Kernel Intrusion Detectors Trustworthy? A Reliability and Calibration Benchmark of QSVM Against Strong Tabular Learners on NSL-KDD"** for consideration as a regular paper in *International Journal of Network Management*.

Quantum Support Vector Machines (QSVM) with the ZZ-feature map have been proposed as a near-term quantum-machine-learning candidate for network intrusion detection (IDS). However, existing studies evaluate them almost exclusively on *accuracy* and only against classical SVM baselines. On tabular IDS data, gradient-boosted trees and random forests usually dominate accuracy, so an accuracy-only comparison misses the question that actually governs deployment: whether the model's probabilistic alerts are *trustworthy*.

To address this gap, we reframe the evaluation around *reliability and calibration*. We benchmark a hardware-constrained four-qubit QSVM against the strongest tabular learners – Random Forest and XGBoost – alongside an SVM-RBF baseline, under a single zero-leakage preprocessing pipeline and a single Platt-scaling protocol, so that observed differences are attributable to the classifier rather than to confounding experimental factors. Using a five-run protocol on NSL-KDD, we report Expected Calibration Error (ECE), Brier score, and AUC-PR with Cohen's d effect sizes.

Our central finding is a *regime-specific reliability* result: on the rare-attack classes (User-to-Root and Remote-to-Local, under 1% of the data and the most security-critical), the quantum kernel is the most trustworthy model, attaining the lowest ECE and Brier score with a large effect size against the tree ensembles. It is also best calibrated at a balanced operating point and in the low-data regime, while remaining only competitive under heavy prior shift and temporal drift – results we report transparently, including where the quantum kernel does not lead. We further show that ranking quality and calibration are distinct (tree ensembles rank well yet are over-confident), and that Platt scaling is the appropriate recalibration tool for margin-based quantum and SVM models but not for tree ensembles.

We believe this manuscript is well aligned with the scope of *International Journal of Network Management*. It provides the first calibration-centric reliability study of a quantum kernel for intrusion detection, together with practical, deployment-oriented guidance on when such a model can be trusted – a topic of growing relevance to readers working on machine learning for network security and on trustworthy AI.

We confirm that:

- This manuscript is original and has not been published previously.
- The manuscript is not currently under consideration by any other journal or conference.
- All authors have reviewed and approved the manuscript and agree with its submission to *International*

Journal of Network Management.

- There are no conflicts of interest related to this work.
- The only self-citation is to our companion manuscript, which is clearly disclosed as such in the reference list.

This manuscript was prepared in LaTeX; as requested for initial submission, we provide a single compiled PDF for review and will supply the full LaTeX source files upon revision.

The corresponding author is **Quan Tran Anh Vo** (votrananhquan1111@gmail.com). We thank you and the reviewers in advance for your time and consideration, and we look forward to your response.

Sincerely,

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(on behalf of all co-authors)
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